

Phase 1 - Environment & Repository Setup

This document captures the environment baseline and repository registration steps performed in Phase 1 of the RHEL server hardening project.

1. System Baseline Snapshot

Five commands were used to capture the starting state of the server before any provisioning or hardening began:

```
cat /etc/os-release
hostnamectl
lscpu
free -h
df -h
```

1.1 OS Release

```
lobnaahmed@servera:~$ cat /etc/os-release
NAME="Red Hat Enterprise Linux"
VERSION="10.2 (Coughlan)"
RELEASE_TYPE=stable
ID="rhel"
ID_LIKE="centos fedora"
VERSION_ID="10.2"
PLATFORM_ID="platform:el10"
PRETTY_NAME="Red Hat Enterprise Linux 10.2 (Coughlan)"
ANSI_COLOR="0;31"
LOGO="fedora-logo-icon"
CPE_NAME="cpe:/o:redhat:enterprise_linux:10.2"
HOME_URL="https://www.redhat.com/"
VENDOR_NAME="Red Hat"
VENDOR_URL="https://www.redhat.com/"
DOCUMENTATION_URL="https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/10"
BUG_REPORT_URL="https://issues.redhat.com/"

REDHAT_BUGZILLA_PRODUCT="Red Hat Enterprise Linux 10"
REDHAT_BUGZILLA_PRODUCT_VERSION=10.2
REDHAT_SUPPORT_PRODUCT="Red Hat Enterprise Linux"
REDHAT_SUPPORT_PRODUCT_VERSION="10.2"
```

1.2 Hostname & System Identity

```
lobnaahmed@servera:~$ hostnamectl
Static hostname: servera.mylab.com
Icon name: computer-vm
Chassis: vm
Machine ID: 4cbd8cbcd7fb443f991ab90abaf25410
Boot ID: 0ef4d94136874e93ba78935fb1b6a8d1
AF_VSOCK_CID: 2096941316
Virtualization: vmware
Operating System: Red Hat Enterprise Linux 10.2 (Coughlan)
CPE OS Name: cpe:/o:redhat:enterprise_linux:10.2
Kernel: Linux 6.12.0-211.7.3.el10_2.x86_64
Architecture: x86_64
Hardware Vendor: VMware, Inc.
Hardware Model: VMWare20.1
Firmware Version: VMW201.00V.21805430.B64.2305221830
Firmware Date: Mon 2023-05-22
Firmware Age: 3y 1month 1w 5d
```

1.3 CPU Architecture

```

lobnaahmed@servera:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          45 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 1
On-line CPU(s) list:    0
Vendor ID:              GenuineIntel
Model name:             Intel(R) Core(TM) i7-6820HQ CPU @ 2.70GHz
CPU family:             6
Model:                 94
Thread(s) per core:     1
Core(s) per socket:     1
Socket(s):              1
Stepping:               3
BogoMIPS:               5424.00

```

1.4 Memory Usage

```

lobnaahmed@servera:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           1.6Gi        1.0Gi        309Mi        9.4Mi        450Mi        597Mi
Swap:          2.0Gi        376Mi        1.6Gi

```

1.5 Disk Usage

```

lobnaahmed@servera:~$ df -Th
Filesystem      Type      Size  Used Avail Use% Mounted on
/dev/mapper/rhel-root xfs       46G   6.6G   39G  15% /
devtmpfs        devtmpfs  805M    0  805M   0% /dev
tmpfs           tmpfs     837M   88K  837M   1% /dev/shm
efivarfs        efivarfs  256K   56K  196K  23% /sys/firmware/efi/efivars
tmpfs           tmpfs     335M   13M  322M   4% /run
tmpfs           tmpfs     1.0M    0   1.0M   0% /run/credentials/systemd-journald.service
/dev/nvme0n1p2   xfs       2.0G  441M   1.6G  23% /boot
/dev/nvme0n1p1   vfat      599M   9.0M  590M   2% /boot/efi
tmpfs           tmpfs     168M   136K  168M   1% /run/user/1000
/dev/sr0         iso9660   11G    11G    0 100% /run/media/lobnaahmed/RHEL-10-2-BaseOS-x86_64

```

2. Red Hat Subscription Registration

The server was registered directly against Red Hat's subscription service rather than relying on a local ISO-based repository. This reflects the standard production path for a licensed, internet-connected system.

2.1 Subscription Status

System Status Details

```
-----
Overall Status: Registered
```

2.2 Repository Configuration

The full `/etc/yum.repos.d/redhat.repo` file is auto-generated by subscription-manager and contains close to one hundred repository definitions covering every Red Hat product line (OpenShift, SAP, JBoss middleware, Container Development Kit, etc.), the vast majority of which are disabled and irrelevant. Only the two repositories actually enabled and used are shown below for clarity.

```

[rhel-10-for-x86_64-baseos-rpms]
name = Red Hat Enterprise Linux 10 for x86_64 - BaseOS (RPMs)
baseurl = https://cdn.redhat.com/content/dist/rhel10/$releasever/x86_64/baseos/os
enabled = 1
gpgcheck = 1

```

```
sslverify = 1

[rhel-10-for-x86_64-appstream-rpms]
name = Red Hat Enterprise Linux 10 for x86_64 - AppStream (RPMs)
baseurl = https://cdn.redhat.com/content/dist/rhel10/$releasever/x86_64/appstream/os
enabled = 1
gpgcheck = 1
sslverify = 1
```

2.3 Active Repository Confirmation

Command run: `dnf repolist`

repo id	repo name
rhel-10-for-x86_64-appstream-rpms	Red Hat Enterprise Linux 10 for x86_64 - AppStream (RPMs)
rhel-10-for-x86_64-baseos-rpms	Red Hat Enterprise Linux 10 for x86_64 - BaseOS (RPMs)

2.4 Storage Inspection

Storage was inspected using `lsblk` and `blkid` against the server's existing NVMe/LVM layout (root and swap logical volumes on a single 50GB NVMe disk). Creating and mounting an additional dedicated data volume was scoped out of this phase due to lab hardware constraints (no spare virtual disk available) and limited LVM experience at the time of the project — noted here as a deliberate scoping decision rather than an oversight.